Solutions — Paper B

Solution 1 (12 marks). This is a bad idea for several compounding reasons. First, Chapter 14 established that instantaneous volatility genuinely varies with time-to-expiry — the Rebonato humped shape — so a single flat number cannot represent the real term structure; every caplet away from the 5-year point would be systematically mispriced, in a direction and magnitude depending on where its own time-to-expiry sits relative to the humped shape's peak. (4 marks.) Second, Chapter 16 showed that even the caplets *within* a single flat-quoted cap have different individual spot vols from the cap's own flat quote — collapsing an entire curve to one cap's flat number discards this distinction twice over. (3 marks.) Third, and most seriously, this single caplet vol curve is the direct input to the Rebonato fit and, downstream, to the entire swaption calibration of Chapters 16–18: an error here does not stay contained to the cap book, it propagates through the whole pipeline and corrupts the swaption fit and every subsequently-priced exotic. (3 marks.) Full marks require naming at least two of these three effects with the correct supporting chapter. (2 marks for overall clarity and correct emphasis that the "savings" are illusory — the stripping step is cheap relative to the damage caused by skipping it.)

Solution 2 (14 marks). This is far more likely to be a calibration degeneracy than a genuine market move. Chapter 8 flagged exactly this pattern for SABR's β/ρ tradeoff, and the same structural issue exists between ρ and ν: both parameters shape the smile (skew and wing width respectively), and different combinations can produce very similar smile shapes over the range of strikes actually quoted in the market, especially if the quoted strike range is narrow relative to the wings the fit is extrapolating into. (6 marks: correctly identifying degeneracy as the more likely explanation, with reference to Chapter 8's β/ρ discussion as the structural analogue.) The decisive evidence here is that *the residual is essentially unchanged* between the two fits: if the market had genuinely moved this much, one would expect the previous day's parameters to now fit noticeably *worse* against today's data, not equally well under two very different parameter sets. (4 marks: correctly using the unchanged residual as the specific evidence.) A concrete check: recompute the implied volatility smile both fits produce, evaluated specifically at the strikes that are actually quoted in the market (not extrapolated wings), and compare them directly — if they agree closely at the quoted strikes and diverge only away from them, this confirms the fits are observationally near-equivalent on the data that matters and the parameter move is an artefact of an under-constrained optimization landscape, not new information. (4 marks: a genuinely concrete, executable check, not a restatement of the diagnosis.)

Solution 3 (14 marks). Chapter 18 states the cascade walks from the *longest*-expiry, shortest-tenor co-terminal swaptions first, proceeding toward the *shortest*-expiry, longest-tenor swaptions last, with each step treating all previously solved parameters as fixed and exact. (5 marks: correctly identifying the cascade's solving order.) The pattern described — excellent fits at long expiry, steadily worsening as expiry shortens inside 3 years — is exactly the signature of the error-accumulation failure mode Chapter 18 flags: since short-expiry swaptions are processed *last* in this ordering, any small pricing or data imprecision introduced early in the cascade has had the most steps to compound by the time the short end is reached, with no mechanism in a pure cascade for a later step to correct an earlier one. (5 marks: correctly connecting the observed pattern to error accumulation specifically, not simply asserting "something is wrong with short expiries.") The recommended remedy is to stop treating the cascade result as final: use it purely as a fast, well-behaved starting guess, and follow it with a joint, regularized optimization across the whole grid simultaneously (Chapter 18), which has the ability to trade off small adjustments across every instrument at once rather than being locked into the cascade's one-directional solving order. (4 marks: correct, specific remedy naming joint optimization as a correction, not merely "recalibrate again.")

Solution 4 (12 marks). The likely cause is that the long-stepped time grid — one step per 3-month tenor period — is adequate for the Bermudan swaption, whose exercise decisions depend only on the rate levels *at* the tenor dates themselves, but is not adequate for an in-arrears caplet, whose payoff genuinely depends on the path of the compounded overnight rate *within* each accrual period, exactly the long-stepping pitfall Chapter 19 describes. (5 marks: correct identification of long-stepping as the cause, with the Bermudan/in-arrears contrast correctly drawn.) Specifically, if the simulation engine effectively treats the in-arrears rate as if it were a single value known at the start of the period (the same simplification implicitly made by a coarse grid that only samples endpoints), it discards the smoothing effect of daily compounding within the period that Chapter 3's worked example showed reduces the rate's effective volatility relative to a single terminal draw — so the simulation overstates the caplet's true volatility, and therefore overstates its price, which matches the observed symptom exactly. (5 marks: correctly explaining *why* the mispricing is specifically an *overstatement*, connecting back to the variance-of-an-average argument from Chapter 3.) The fix is not to abandon long-stepping for the Bermudan (where it is appropriate and efficient) but to use a finer time grid, or an explicit in-arrears convexity adjustment, specifically for products whose payoff depends on intra-period behaviour. (2 marks for a correctly scoped remedy that does not overcorrect by proposing fine-stepping everywhere.)

Solution 5 (12 marks). Two distinct reasons, each worth 6 marks for a correctly explained, chapter-referenced failure mode. *First*, cascade-only calibration is vulnerable to the error accumulation described in Chapter 18 and tested directly in Question 3 above: a cascade that "looks fine" by eye on a spot check of a few instruments can still be silently compounding error toward one end of the grid, and without a joint refinement step there is no mechanism to catch or correct this — "looked basically fine" is precisely the kind of impression a degrading-but-not-obviously-broken cascade produces. *Second*, a cascade calibration run in isolation, with no regularization toward the previous day's fit or a historical prior, has no defence against the ill-posedness problem of Chapter 18: on a day with only modest genuine market movement, an unregularized fit can still land on a meaningfully different point in the (flat, under-determined) solution space purely due to small quoting noise, producing spurious day-over-day jumps in calibrated parameters, and therefore in computed Greeks and risk numbers, that do not reflect any real market move — a serious problem for both P&L attribution and risk management even when the raw fit quality "looks fine" on the day it is produced.

Solution 6 (16 marks). A structured, time-boxed triage, drawing on Chapter 20's framework: *(i)* Because the failure is isolated to a single instrument rather than spread across the grid, start from the strong prior that this is a data problem, not a model failure (2 marks). *(ii)* Pull the raw vendor quote for this specific $3y \times 7y$ receiver and compare it against yesterday's value and, if available, a second independent data source (3 marks). *(iii)* Check it against neighbouring grid points — adjacent expiries and tenors on the swaption cube, and the corresponding payer quote via swaption put–call-style consistency — since a genuine, isolated data error typically shows a sharp discontinuity against its neighbours that a real market move would not (3 marks). *(iv)* If the quote is confirmed bad, correct or exclude it and re-run only the affected downstream stage rather than the full pipeline, to stay within the available time (3 marks). *(v)* If the quote checks out as genuine, consider whether this specific point sits in a region where the frozen-weights approximation (Chapter 17) is known to degrade, or flag it as a real, isolated market dislocation worth relaying to the desk rather than silently absorbing into the fit (3 marks). *(vi)* If the question remains unresolved by market open, the correct action is to invoke the kill-switch: do not publish a calibration that fails its own validation tolerance, fall back to the previous day's calibration (clearly flagged as stale) for pricing that continues to depend on this segment of the cube, and escalate to the model risk function and desk head rather than either silently overriding the failed check or delaying the desk's open (2 marks). Full marks require the steps to be both correctly ordered (cheapest, most likely diagnosis first) and time-aware, not simply a list of unordered checks.

Solution 7 (10 marks; sample five-sentence answer, full marks for any answer covering the same substance in plain, non-technical language). "Historically, almost all of the day-to-day movement in the entire interest rate curve can be described by just three simple patterns: the whole curve shifting up or down together, the curve steepening or flattening, and the curve bending in the middle. Because these three patterns explain the vast majority of how rates actually move, a model built around just three of these 'movement patterns' captures nearly everything that matters for most products, at a fraction of the computational cost of tracking every point on the curve independently. However, some products care specifically about whether two very close points on the curve — say, a 9-year and a 9.25-year rate — move slightly *differently* from each other, and that kind of fine, localized wiggle is exactly what gets smoothed away when we compress the whole curve down to just three patterns. For those specific products, we need the fuller, more detailed picture of how every point relates to every other point, not the compressed version. So the three-pattern simplification is a good general-purpose choice, but not a universal one — it depends on what the specific product actually cares about."

Solution 8 (10 marks). With only three swaption tenors and two cap maturities quoted, several stages of the Chapters 16–18 pipeline degrade materially. Caplet stripping (Chapter 16) can only recover very coarse volatility buckets — with just two cap maturities, at most two independent spot-vol data points are available, nowhere near enough to meaningfully constrain the Rebonato hump's four parameters, particularly the hump location and decay rate (b, c), which need several maturities spanning the hump to identify at all. (3 marks.) The swaption side is worse: Chapter 17–18's joint calibration needs enough swaption coverage across expiry and tenor to identify a full correlation structure, and three tenor points is very likely insufficient to pin down even a two-parameter exponential-decay correlation form reliably, let alone anything richer — the ill-posedness problem from Chapter 18 becomes far more severe, not merely present, under this degree of data scarcity. (4 marks.) A practical mitigation is to borrow structural information from a more liquid, economically related currency or market — using its fitted volatility-hump shape and correlation-decay parameters as a strong regularization prior (or even holding shape parameters fixed and calibrating only a small number of level/scale parameters to the thin local data) — so the model relies on genuinely observed local data only for what that data can actually identify, and leans on external structural assumptions, clearly documented as such, for everything else. (3 marks: a concrete, specific mitigation, not merely "get more data," which is correct but not actionable given the stated constraint.)